

ASSIGNMENT SUBMISSION

Web Development Fundamentals

1. Frontend vs. Backend vs. Full-Stack

Frontend (Client-Side): Everything users see and interact with (buttons, colors, layouts). Built using HTML, CSS, and JavaScript.

Backend (Server-Side): The "brain" behind the scenes. It handles data processing, databases, and server logic. Built with Node.js, Python, or Java.

Full-Stack: A developer who can work on both the Frontend and Backend layers of an application.

***Real-World Example (Amazon):** The search bar and product images you see are **Frontend**. When you click 'Buy' and the system checks your credit card and updates inventory, that is the **Backend**.*

2. Client-Server Model Diagram



3. How a Browser Requests a Page

1. **DNS Lookup:** Browser converts the URL (e.g., google.com) into an IP address.
2. **HTTP Request:** Browser sends a "GET" request to that IP address.
3. **Server Processing:** The server finds the requested file and sends it back.
4. **Rendering:** The browser receives HTML, fetches CSS/JS, and "paints" the pixels on your screen.

4. Web Development Environment Tools

Tool	Purpose
Code Editor (VS Code)	Where you write and organize your source code.
Web Browser (Chrome)	Used to test and debug the output of your code.
Version Control (Git)	Tracks changes and allows collaboration with others.
Package Manager (NPM)	Installs external libraries and frameworks.

5. Web Servers

A web server is software (or hardware) that stores web files and delivers them to users via HTTP. It "serves" the content you request.

- **Apache:** One of the oldest and most widely used servers.
- **Nginx:** Known for high performance and reverse proxying.
- **Microsoft IIS:** Used primarily for Windows-based hosting.

6. Roles in a Project

- **Frontend Developer:** Focuses on UI/UX and user interaction.
- **Backend Developer:** Focuses on logic, security, and API integration.
- **Database Administrator (DBA):** Focuses on data integrity, optimization, and storage management.

7. VS Code Configuration

To set up VS Code for web dev: 1. Install **Live Server** extension. 2. Install **Prettier** for formatting. 3. Set up Emmet for fast HTML coding

8. Static vs. Dynamic Websites

Static: Content is fixed. Every user sees the same thing (e.g., a simple portfolio). Fast but hard to update.

Dynamic: Content changes based on user data or time (e.g., Facebook News Feed). Uses databases to generate pages on the fly.

9. Web Browsers s Rendering Engines

- **Google Chrome / Edge:** Uses **Blink** engine (fast, widely supported).
- **Safari:** Uses **WebKit** (optimized for Apple hardware).
- **Firefox:** Uses **Gecko** (privacy-focused and open-source).
- **Opera:** Also uses **Blink** (based on Chromium).

10. Advanced Web Architecture Flow

